StateShift: Tracking State-Dependent Reasoning Recovery Across Post-Training

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Abstract

Standard evaluations of reinforcement-learning post-training in large language models emphasize aggregate benchmark accuracy, potentially obscuring changes conditional on local reasoning state. We introduce ***StateShift****, a framework for measuring target-transition recovery under matched reasoning states. In a controlled study (N=454, 29,056 rollouts), we observe an 11.76-percentage-point state-by-checkpoint interaction between the base and step-256 checkpoints (Γ_{256}=0.1176, 95% problem-blocked bootstrap CI [0.0955, 0.1400]). Across nine empirically evaluated checkpoints, the interaction was consistent with a non-decreasing trajectory under prespecified order-restricted analysis despite local variation in unconstrained estimates, and was already detectable at the earliest available post-training checkpoint, t=32 (Γ_{32}=0.0333, multiplicity-adjusted 95% CI [0.0011, 0.0655]). In a separate analysis of 3,200 unperturbed rollouts, 18.19% contained a verifier-confirmed natural reasoning error; among 582 qualifying error episodes, 180 subsequently satisfied the prespecified autonomous recovery criterion (30.93%, 95% problem-blocked bootstrap CI [27.19%, 34.82%]). These results show that state-conditioned evaluation exposes behavioral structure not captured by aggregate correctness alone.*

1. Introduction

Reinforcement learning from verifier feedback (RLVR) has emerged as a dominant paradigm for incentivizing mathematical reasoning capabilities in large language models. However, standard evaluation protocols rely almost exclusively on aggregate benchmark accuracy, providing limited visibility into how post-training alters local reasoning dynamics conditional on intermediate state. We introduce **StateShift** to evaluate state-conditioned target-transition recovery.

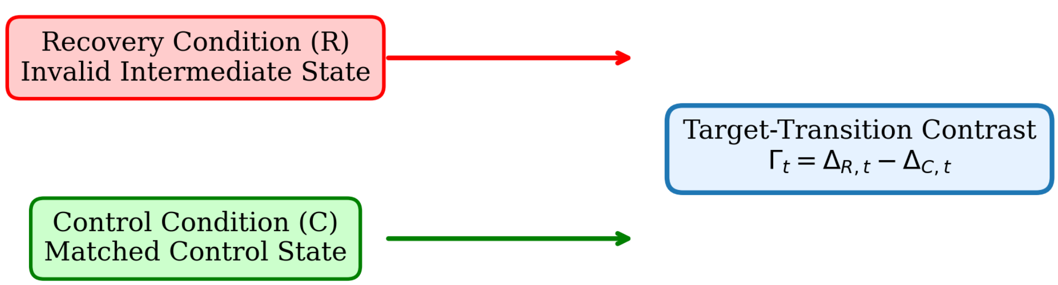

Figure 1: The StateShift experimental framework evaluating matched Recovery and Control states.

2. Primary Controlled Study Results

Between the base and step-256 checkpoints, we observe an 11.76-percentage-point state-by-checkpoint interaction (Γ_{256} = +0.1176, p < 0.0001, 95% bootstrap CI [0.0955, 0.1400]).

Condition	Base (t=0)	Step 256 (t=256)	Change (Delta)	Interaction (Gamma_256)	95% Bootstrap CI
Recovery (R)	0.3834	0.7039	+0.3205	--	--
Control (C)	0.3892	0.5921	+0.2029	--	--
Interaction	--	--	--	+0.1176	[+0.0955, +0.1400]

3. Nine-Checkpoint Empirical Trajectory

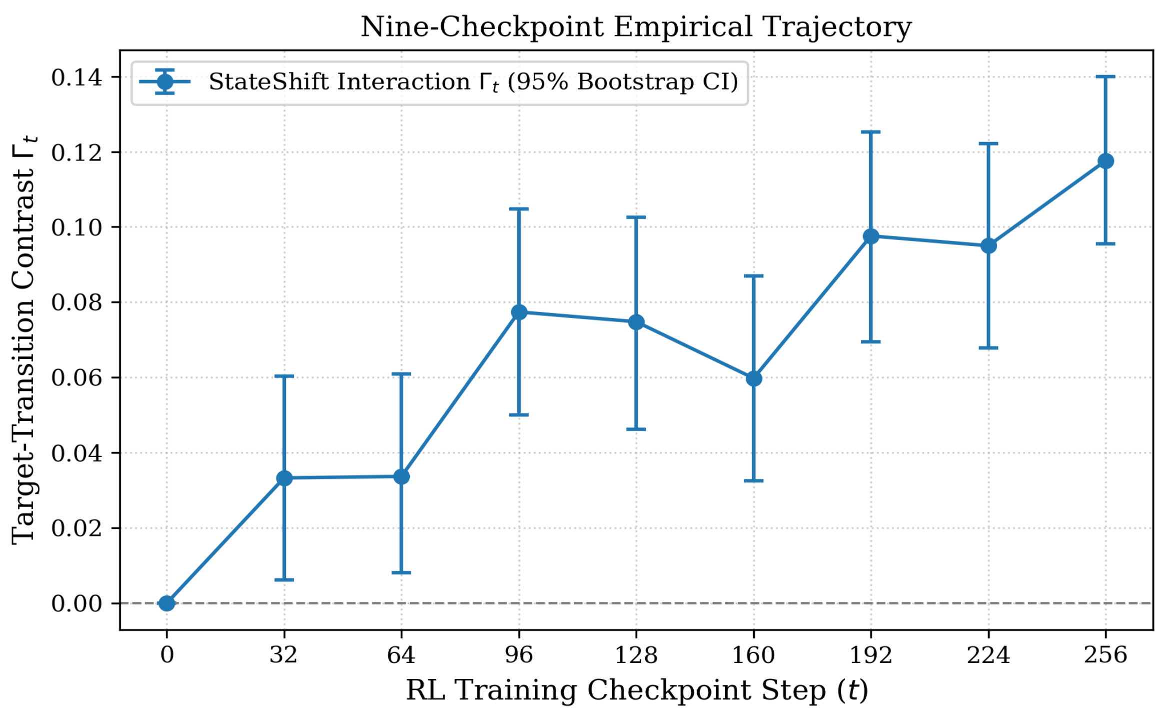

Figure 2: Full nine-checkpoint empirical interaction trajectory with 95% bootstrap CIs.

4. Natural Post-Error Recovery

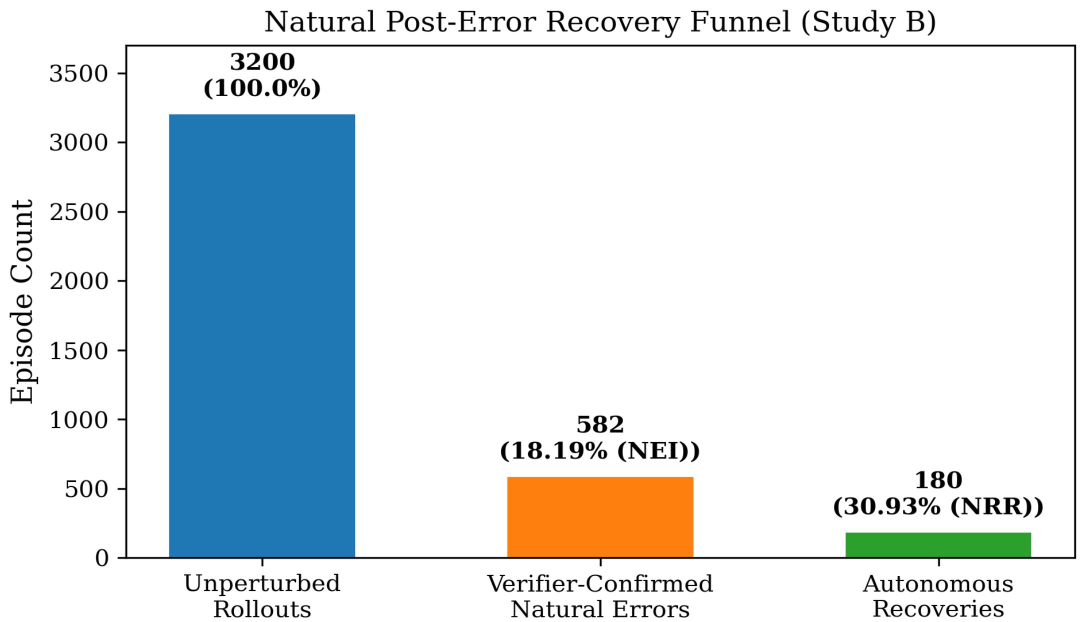

Figure 4: Natural post-error recovery funnel (3,200 rollouts -> 582 errors -> 180 recoveries).